

Intergenerational Housing Misallocation and Fertility

Tommaso De Santo

Abstract

Children increase a household's demand for living space. Yet large rental units are scarce, buying a family-sized home requires liquid wealth and access to credit, and older households own much of the stock of large homes. I study whether this allocation of housing raises the cost of having children for young households. I first develop a simple overlapping-generations model in which rental limits, down-payment constraints, and the retention decisions of older owners determine who occupies family-sized housing. I then study the same forces in a finite-life model with income risk, saving, tenure choice, transaction costs, and bequests. The model distinguishes changes in family size among current residents from changes in local births due to entry into the housing market. The quantitative model suggests that restricted access to family-sized housing matters for households deciding whether and when to have children, and that home purchases and births are closely connected near the end of the fertile years.

Status of this draft

- I study a two-generation model that links access to family-sized housing to the cost of children, derives the exact marginal-value gap between constrained young buyers and old incumbents, defines equilibrium as a transition between positive steady states, and gives explicit conditional existence results.
- I build a finite-life quantitative model with income risk, saving, fertility, tenure choice, transaction costs, and bequests. The solution produces a close connection between home purchases and births and places a high value on additional space among constrained renters.
- A grant for purchasing a family-sized home raises total births by approximately 2.1 percent and the housing price by 0.90 percent. Combining the grant with a tax on large homes raises total births by approximately 2.7 percent and lowers the housing price by 18.4 percent.
- The analytical model characterizes mixed renter-owner positive steady states, gives conditional existence results for a perfect-foresight transition, and computes a horizon-stable transition approximation after one permanent fertility-preference change. It also derives the exact local surplus from reallocating housing across generations. The global ranking of policies that change population remains a separate normative choice rather than an unstated welfare criterion.
- The quantitative model requires further evaluation, and further work is needed on the calibration: the income-risk variance is provisional, and the model overstates late-life homeownership. The current version makes a one-time choice of completed fertility, so all children arrive in a single cohort and age together. The main next extension is to add sequential fertility, alongside strengthening the old-age downsizing margin, reassessing identification, and further validating the entry response in the policy exercises.

- I do not report the project’s empirical analysis in this note. Empirical moments from the CPS, ACS, and PSID are used to discipline and evaluate the quantitative model, but the construction of those moments and the reduced-form evidence are left for a subsequent version.

1 Introduction

Children increase a household’s demand for living space. Yet family-sized housing is unevenly allocated across generations. Older households own much of the stock of large homes, including after their children leave, while young households deciding whether to have children are more likely to rent and have limited liquid wealth. Large rental units are scarce, and buying a family-sized home requires access to credit and a substantial down payment. This note asks whether this allocation of housing across generations raises the cost of having children and affects completed fertility—the number of children a household has by the end of its fertile years—and local births.

The mechanism operates through access to space. When the rental market does not offer sufficiently large units, a constrained renter values an additional room above its posted rent. Ownership provides access to larger homes, but a young household may be unable to finance the required down payment. At the same time, transaction costs and tax rules can encourage older owners to remain in large homes after their children leave. The resulting scarcity is therefore not only a question of the aggregate housing stock. It also concerns who occupies family-sized homes and whether those homes reach households making fertility decisions.

I study this mechanism in two steps. A simple overlapping-generations model shows how rental limits, financing constraints, and old-owner retention raise the effective cost of accommodating children. I then embed these forces in a finite-life model with income risk, saving, fertility, tenure choice, transaction costs, and bequests. The quantitative model finds that many renters making family-size decisions are constrained by the rental-size limit and that home purchases and births are closely connected near the end of the fertile years. I compare a grant for purchasing family-sized homes with a package that combines the grant with a tax on large homes. Allowing entry and population to respond, the grant raises total births by approximately 2.1 percent and the housing price by 0.90 percent; the package raises total births by approximately 2.7 percent and lowers the price by 18.4 percent.

The distinction between birth timing and completed fertility is central to this note. Quantitative lifecycle models show how economic conditions can affect both when households have children and their completed fertility (Sommer, 2016; Doepke and Kindermann, 2019). Empirical work shows that house prices, housing wealth, and mortgage access affect childbearing (Lovenheim and Mumford, 2013; Dettling and Kearney, 2014; Hacamo, 2021). Recent studies also examine longer-run outcomes: Dettling and Kearney (2025) present supplementary cohort correlations—positive for White women—between cumulative FHA/VA lending and children ever born, and van Doornik et al. (2025) use random variation in the timing of Brazilian housing-credit lotteries to estimate the effects of earlier access on completed fertility among participants observed after age 40. These findings provide the empirical motivation for a quantitative lifecycle model that separates changes in birth timing from changes in completed fertility and can be used to evaluate housing policies in equilibrium.

2 Analytical Model

This section isolates two implications of housing access in a two-period overlapping-generations economy. First, a financing-constrained young buyer can value family-sized space more than an old incumbent who would realize a taxable gain by selling it. Second, a permanent change in desired fertility can move the economy between positive steady states, so house-price appreciation and taxable gains can arise along the transition even though they vanish at both endpoints. Appendix A provides the household solutions, conditional transition results, and a high-accuracy mixed-tenure transition approximation.

2.1 Environment and household choices

At date t , the economy contains Y_t young households and O_t old households. A young household has income y_i , liquid wealth b_i , and chooses consumption c , housing h , completed fertility n , saving a' , and tenure $m \in \{R, O\}$. The pair (y_i, b_i) is drawn from a fixed distribution F in each cohort; estates are warm-glow expenditures rather than the source of the next cohort's b_i . A child requires $\chi > 0$ units of goods and $\kappa > 0$ units of space. Young and old flow utilities are

$$u_t^Y(c, h, n) = \log c + \alpha \log(h - \kappa n) + \vartheta_t \log n, \quad (1)$$

$$u_t^O(c^O, h^O, e) = \log c^O + \gamma \log h^O + \omega_B \log e, \quad (2)$$

where e is a warm-glow estate. A child born at t produces $\nu > 0$ young households at $t + 1$, after survival and household formation. Thus $Y_{t+1} = \nu \bar{n}_t Y_t$ and $O_{t+1} = Y_t$, where $\bar{n}_t$ is average fertility among the young.

The fixed housing stock is $\bar{H}$. Rental units satisfy $h \leq \bar{h}^R$, owner units satisfy $h \leq \bar{h}^O$, and $\bar{h}^R < \bar{h}^O$. A one-period bond has price $q \in (0, 1)$ and gross return $R = q^{-1}$. Competitive housing intermediation implies the rental user cost

$$r_t = (1 + \tau^p)P_t - qP_{t+1}, \quad \ell_t = \tau^g[P_t - P_{t-1}]_+, \quad (3)$$

where P_t is the house price, τ^p the property-tax rate, and ℓ_t the capital-gains tax per unit sold by an old owner. Housing retained until death receives a stepped-up basis; intermediaries mark their basis to market. A young buyer finances the share $\phi \in (0, 1)$ and therefore posts $(1 - \phi)P_t h$ at purchase.

Conditional on renting, the young household pays $r_t h$ and enters old age with financial wealth Ra' . Conditional on owning, it solves

$$\begin{aligned} W_t^O(i) &= \max_{c, h, n, a'} u_t^Y(c, h, n) + \beta V_{t+1}^O(a_{t+1}, h) \\ \text{s.t. } &c + \chi n + a' + [(1 - \phi) + \tau^p]P_t h = y_i + b_i + T_t, \\ &a_{t+1} = Ra' - R\phi P_t h, \quad (1 - \phi)P_t h \leq b_i, \quad h \leq \bar{h}^O, \quad a' \geq 0, \end{aligned} \quad (4)$$

and the renter problem $W_t^R(i)$ replaces the two owner cash-flow equations with $c + \chi n + a' + r_t h = y_i + b_i + T_t$, $h \leq \bar{h}^R$, and $a' \geq 0$. Only b_i is pledgeable at closing. The transfer T_t is the equal per-household rebate of housing-tax revenue.

An old renter spends $a + T_t$ on c^O , rental housing, and the estate. An old incumbent enters with financial wealth a and inherited housing H and has the flow-equivalent budget

$$c^O + qe + (r_t - \ell_t)h^O = a + (P_t - \ell_t)H + T_t, \quad 0 < h^O \leq H, \quad e \geq P_{t+1}h^O. \quad (5)$$

The marginal cost of retaining a unit is therefore $r_t - \ell_t$: selling realizes the current tax, while retaining the unit avoids it.

To keep both tenures active without altering conditional policies, households receive iid type-I extreme-value tenure tastes with scale $\kappa_T > 0$ and an ownership intercept $\bar{\xi}$. The owner share for type i is

$$\pi_t^O(i) = \frac{\exp\{[W_t^O(i) + \bar{\xi}]/\kappa_T\}}{\exp\{W_t^R(i)/\kappa_T\} + \exp\{[W_t^O(i) + \bar{\xi}]/\kappa_T\}}. \quad (6)$$

The taste shocks smooth aggregate tenure choice; they do not enter the conditional renter or owner first-order conditions.

2.2 Housing access and conditional policies

Let $A = 1 + \gamma + \omega_B$, $k = \beta A$, and define lifetime resources and tenure-specific user costs by

$$\mathcal{W}_{it} = y_i + b_i + T_t + qT_{t+1}, \quad r_t^R = r_t, \quad r_t^O = r_t + q\ell_{t+1}. \quad (7)$$

When young saving and next-period old choices are interior, the Euler equation reduces each conditional problem to

$$(1 + k)c + \chi n + r_t^m h = \mathcal{W}_{it}. \quad (8)$$

Writing $B_t = 1 + k + \alpha + \vartheta_t$, unconstrained policies are

$$c_t^m = \frac{\mathcal{W}_{it}}{B_t}, \quad n_t^m = \frac{\vartheta_t \mathcal{W}_{it}}{B_t(\chi + \kappa r_t^m)}, \quad h_t^m = \frac{\alpha \mathcal{W}_{it}}{B_t r_t^m} + \kappa n_t^m. \quad (9)$$

The effective housing limits are

$$\hat{h}_{it}^R = \bar{h}^R, \quad \hat{h}_{it}^O = \min \left\{ \bar{h}^O, \frac{b_i}{(1 - \phi)P_t} \right\}. \quad (10)$$

If the unconstrained policy exceeds $\hat{h}_{it}^m$, housing equals the limit and fertility is the unique interior root

$$\frac{\vartheta_t}{n} - \frac{\alpha \kappa}{\hat{h}_{it}^m - \kappa n} - \frac{(1 + k)\chi}{\mathcal{W}_{it} - r_t^m \hat{h}_{it}^m - \chi n} = 0. \quad (11)$$

The housing first-order condition prices the access constraint in goods:

$$\frac{\alpha c_t^m}{h_t^m - \kappa n_t^m} = r_t^m + \zeta_{it}^m, \quad \zeta_{it}^{O,F} = \frac{\mu_{it}}{\lambda_{it}}(1 - \phi)P_t, \quad (12)$$

where λ_{it} is the marginal value of current resources and μ_{it} is the multiplier on the down-payment constraint. When the owner-size limit is slack, $\zeta_{it}^{O,F}$ is the entire owner access wedge.

2.3 Positive steady states and transitions

Let $\mathcal{N}(P, T; \vartheta)$ be average fertility and let $\mathcal{H}^Y(P, T; \vartheta)$ and $\mathcal{H}^O(P, T; \vartheta)$ be housing per young and old household after tenure choice and the induced old-state distribution. Write $\mathcal{H} = \mathcal{H}^Y + \mathcal{H}^O$. Market clearing and the government budget are

$$Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}, \quad (13)$$

$$(Y_t + O_t)T_t = \tau^p P_t \bar{H} + \ell_t O_t \bar{s}_t^O, \quad (14)$$

where $\bar{s}_t^O$ is housing sold by old owners per old household, assigning zero sales to renters.

Proposition 1 (Positive steady state). *Every positive steady state is a root $(P^*, T^*) \in \mathbb{R}_{++}^2$ of*

$$\nu \mathcal{N}(P, T; \vartheta) - 1 = 0, \quad T - \frac{\tau^p P}{2} \mathcal{H}(P, T; \vartheta) = 0. \quad (15)$$

Given a root, $Y^ = O^* = \bar{H}/\mathcal{H}(P^*, T^*; \vartheta)$ and $\ell^* = 0$. Hence every positive steady state has average fertility $1/\nu$, but prices, transfers, tenure, individual fertility, housing, and population scale can differ across steady states.*

Existence follows under the boundary and continuity conditions in Appendix A.3; local uniqueness requires the two-by-two Jacobian of (15) to be nonsingular. The steady state is thus an equilibrium implication, not a normalization. Replacement fixes average fertility only after imposing stationarity.

Suppose ϑ_t rises permanently at date zero and the economy starts at the old positive steady state. The predetermined state is $S_t = (Y_t, O_t, G_t, P_{t-1})$, where G_t is the distribution of old financial wealth, inherited housing, and tenure. Given a candidate perfect-foresight sequence, household policies generate the two date- t residuals

$$\mathcal{R}_t^H = Y_t \bar{h}_t^Y + O_t \bar{h}_t^O - \bar{H}, \quad (16)$$

$$\mathcal{R}_t^G = (Y_t + O_t)T_t - \tau^p P_t \bar{H} - \ell_t O_t \bar{s}_t^O. \quad (17)$$

A transition equilibrium is a root of this infinite sequence together with the cohort and distribution laws, converging to the new steady state. Computation uses terminally closed finite-horizon truncations: prices, transfers, and value functions are fixed at the new steady state beyond the terminal date, while the terminal endogenous state is allowed to approach it. A regime-consistent price branch and a nonsingular stacked Jacobian give a locally unique truncation for sufficiently small permanent shocks; explicit boundary signs give global existence of a truncation. A convergent sequence of such roots is an exact infinite-horizon transition under the uniform-tail condition in Appendix A.4.

The capital-gains basis needs no second shock. If equilibrium prices satisfy $P_s > P_{s-1}$ on several successive dates, then $\ell_s = \tau^g(P_s - P_{s-1}) > 0$ for each corresponding old cohort. A two-period life makes the preceding date's price the basis; it does not restrict the number of dates over which the equilibrium price adjusts.

2.4 Intergenerational reallocation and fertility

Consider an old incumbent at an interior retention choice and a young owner whose down-payment constraint binds while the owner-size limit is slack. Their marginal values of one housing unit, in current goods, are

$$MV_{it}^Y = r_t + q\ell_{t+1} + \zeta_{it}^{O,F}, \quad MV_{jt}^O = r_t - \ell_t. \quad (18)$$

Proposition 2 (Conditional intergenerational reallocation). *Fix fertility, cohort masses, tenure, and all unaffected household allocations. Suppose a planner can transfer the marginal unit's title and occupancy, provide pledgeable bridge finance, compensate the incumbent with lump-sum transfers, and reclaim the incremental gains-tax rebates at dates t and $t+1$. If the real transfer cost is K_{ijt} , moving a sufficiently small unit from old incumbent j to constrained young buyer i is a compensated improvement when*

$$\underbrace{\zeta_{it}^{O,F} + \ell_t + q\ell_{t+1}}_{MV_{it}^Y - MV_{jt}^O} > K_{ijt}. \quad (19)$$

At a steady state the condition reduces to $\zeta_i^{O,F} > K_{ij}$.

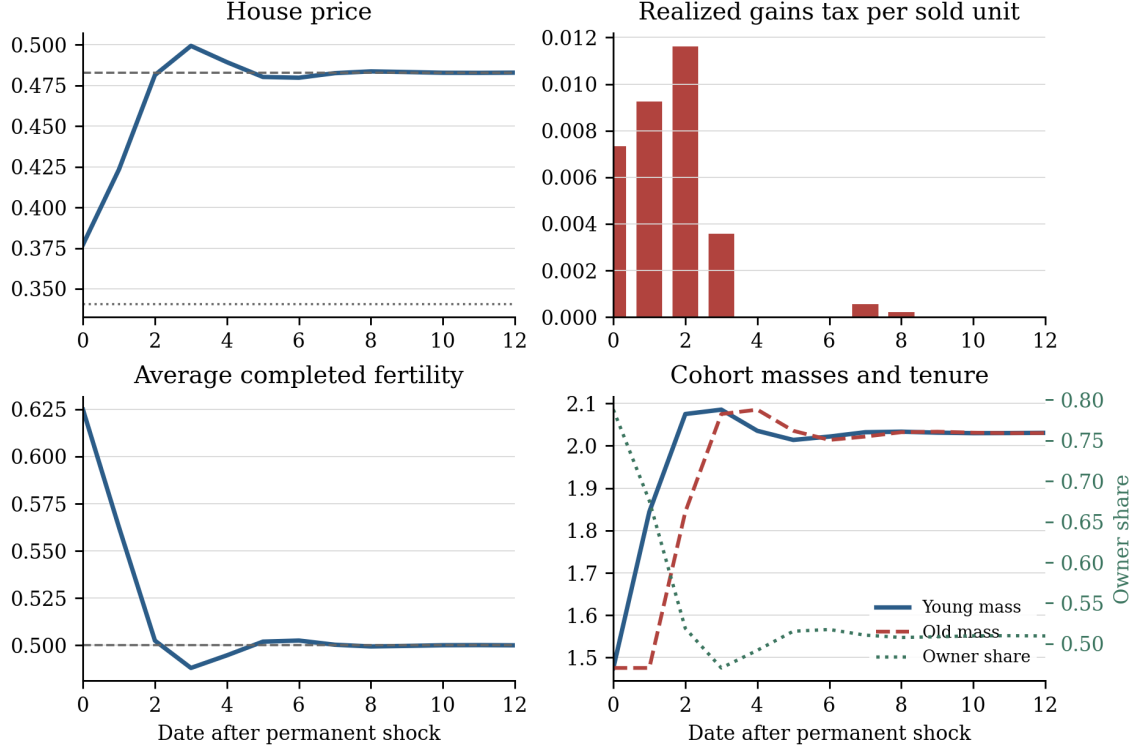


Figure 1: A terminally closed mixed-tenure transition approximation after one permanent increase in the fertility preference. Dotted and dashed lines in the price panel are the old and new steady-state prices; the horizontal fertility line is replacement $1/\nu$. The model has positive renter and owner masses on every displayed date. The preference change is the only exogenous shock.

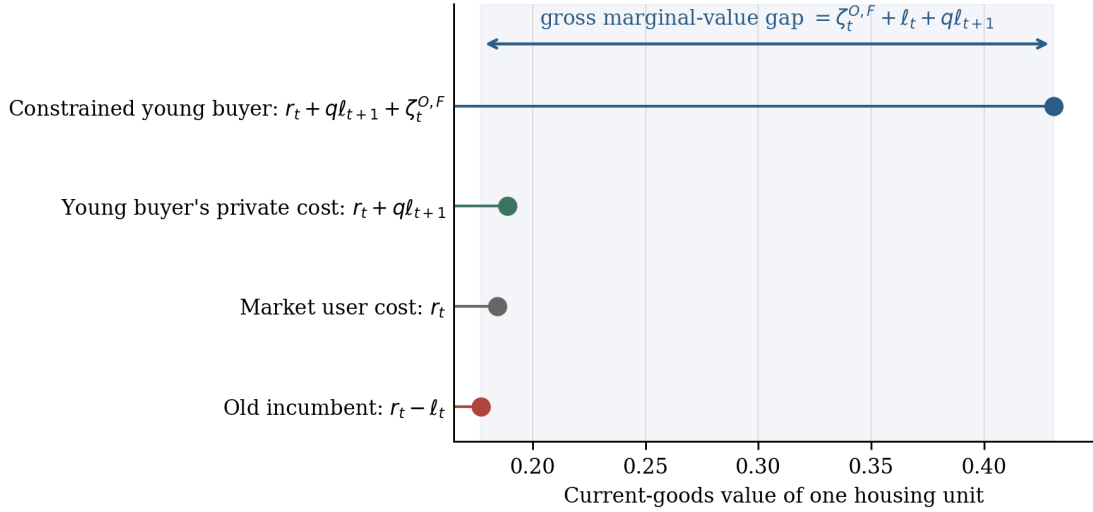


Figure 2: Marginal values on the computed path. The financing wedge raises the young buyer's value; realization taxes lower the old incumbent's value and raise the buyer's anticipated private cost. The displayed object is the gross marginal-value gap. Under the ledger in Proposition 2, a compensated improvement obtains when this gap exceeds the real transfer cost.

This is deliberately a local allocation result. A reform that changes fertility changes the number of future households, so a global ranking also requires an explicit social criterion for comparing populations. The reallocation proposition instead fixes population and identifies the exact intergenerational wedge before choosing such a criterion.

Housing access nevertheless has a precise fertility implication. Holding tenure, prices, lifetime wealth, and the active set fixed, let $D = \mathcal{W}_{it} - r_t^m \hat{h} - \chi n$ and $S = \hat{h} - \kappa n$. Differentiating (11) gives

$$\frac{dn}{d\hat{h}} = \frac{\alpha\kappa/S^2 - (1+k)\chi r_t^m/D^2}{\vartheta_t/n^2 + (1+k)\chi^2/D^2 + \alpha\kappa^2/S^2}. \quad (20)$$

The denominator is positive. Using the housing first-order condition, the exact sign restriction is

$$\frac{dn}{d\hat{h}} \geq 0 \iff \chi \leq \frac{(1+k)\kappa(r_t^m + \zeta_{it}^m)^2}{\alpha r_t^m}. \quad (21)$$

More accessible space raises fertility when children are sufficiently space-intensive relative to their goods cost. This partial-equilibrium sign does not by itself sign a funded policy: prices, rebates, tenure, and lifetime resources also respond in general equilibrium.

3 Quantitative Lifecycle Model

The quantitative model extends the analytical framework to a finite lifecycle. It preserves the central mechanism: children raise the demand for space, large rental units are scarce, ownership provides access to larger homes but requires a down payment, and older households may retain family-sized homes after their children leave. The model continues to have one housing market so that the allocation of space across generations, rather than spatial sorting, remains the central object.

The quantitative model adds the margins needed to evaluate this mechanism. Households face income risk, save toward homeownership, and adjust their housing over the lifecycle. Owner housing is a discrete ladder, while renters choose space continuously up to a size limit. Households choose completed fertility discretely during the fertile years; children subsequently age and leave the household, while completed fertility continues to affect bequests. Housing supply is upward sloping, and entry can change the scale of the market in counterfactuals. Older households remain in their homes because selling is costly and because housing enters the estate.¹

3.1 Lifecycle and Income

Time is discrete. Households enter adult life at age $a = 1$, retire after age J_R , and die at the end of age J . Before retirement, labor income is the product of a common wage, a deterministic age profile, and a persistent idiosyncratic component:

$$y(a, z) = \begin{cases} (1 - \tau_{\text{pay}})w e_a z, & a \leq J_R, \\ p^{\text{ret}}, & a > J_R, \end{cases} \quad z' \sim \Pi^z(\cdot|z), \quad (22)$$

Here, e_a is the deterministic profile of expected household labor earnings over the lifecycle, normalized to average one over working ages. The component z captures persistent differences in labor earnings across households of the same age and is normalized to have invariant mean one. The common wage

¹For now, the quantitative exercises do not study capital-gains taxation or stepped-up basis.

w therefore sets the overall earnings scale, e_a governs the common lifecycle profile, and z governs within-age earnings risk.

I specify $\log z$ as an annual AR(1), sample the implied process every four years, and approximate it by a five-state Rouwenhorst chain. Annual persistence is 0.960 and the innovation standard deviation is 0.0645, implying four-year persistence of 0.85 while preserving the stationary log-earnings dispersion of the annual process.

3.2 Preferences and Bequests

Children raise minimum consumption and housing needs while they live at home. Let n_s be the number of dependent children and let n denote completed fertility; the two differ after children mature and leave. Household needs are

$$\begin{aligned}\bar{c}(n, s) &= \bar{c}_0 + \bar{c}_n n_s, \\ \bar{h}(n, s) &= \bar{h}_0 + \bar{h}_{\text{jump}} \mathbf{1}\{n_s \geq 1\} + \bar{h}_n n_s.\end{aligned}\tag{23}$$

The first child can create a discrete housing-space jump; additional dependent children raise the required space linearly.

Preferences are CRRA over a Stone–Geary composite of consumption and usable housing space. Write h^R for rented space and h for the owned housing stock; only one can be positive. The owner service premium applies after household needs are removed:

$$u(c, h^R, h; n, s) = \frac{[(c - \bar{c}(n, s))^{\alpha_c} \mathbf{h}(h^R, h; n, s)^{1-\alpha_c}]^{1-\sigma}}{1 - \sigma} + \psi_{\text{child}} n_s,\tag{24}$$

where

$$\mathbf{h}(h^R, h; n, s) = \begin{cases} h^R - \bar{h}(n, s), & h = 0, \\ \chi_O [h - \bar{h}(n, s)], & h > 0, \end{cases} \quad \chi_O > 0.\tag{25}$$

When $\chi_O > 1$, ownership provides more utility from each unit of space left after household needs are met. It does not change the physical number of rooms required by children or the quantity of housing counted in market demand.

At death, households receive warm-glow bequest utility. The house passes to the estate at market value: there is no forced sale, no transaction cost, and no tax at death, so bequeathable wealth is $W^B = b + Ph$, liquid wealth plus the gross value of any house held. For $\sigma \neq 1$, bequest utility is

$$\mathcal{B}(W^B, n) = \theta_0 (1 + \theta_n n) \frac{(\theta_1 + \max\{W^B, 0\})^{1-\sigma} - \theta_1^{1-\sigma}}{1 - \sigma}.\tag{26}$$

For $\sigma = 1$, the fraction is $\log(\theta_1 + \max\{W^B, 0\}) - \log \theta_1$. The normalization sets $\mathcal{B}(0, n) = 0$, while the multiplicative term makes completed fertility scale the marginal value of a positive estate. Completed fertility can therefore affect saving even after children have left the household.

3.3 Housing, Finance, and Taxes

There is one aggregate housing-services market. The rental user cost r and asset price P satisfy

$$r = (q^{-1} - 1 + \delta + \tau^p)P.\tag{27}$$

Housing supply is upward sloping:

$$H^S(r) = \bar{H} \left(\frac{r}{\bar{r}} \right)^\eta. \quad (28)$$

Here $\bar{r}$ is the reference rental user cost, so housing supply equals $\bar{H}$ at $r = \bar{r}$. The fixed-stock compact model is the limiting case $\eta = 0$ around the benchmark stock. More generally, η determines whether an increase in housing demand appears mainly as a higher user cost or as additional housing.

Households either rent housing services or own one of K discrete housing sizes. The owned housing stock h belongs to

$$\mathcal{H} = \{0, H_1, \dots, H_K\}, \quad (29)$$

where $h = 0$ denotes a renter and $h = H_k$ an owner of a house of size H_k , ordered so that $H_1 < \dots < H_K$. Thus tenure is implicit in the housing position rather than carried by a separate indicator. Renters choose h^R continuously up to $\bar{h}^R$; owners choose from the discrete ladder. The ladder is the quantitative version of the family-housing segmentation in Section 2.

Renters choose $h^R \in [0, \bar{h}^R]$ and cannot borrow. Owners choose from the discrete ladder and can borrow against the house. They pay maintenance and property taxes through $(\delta + \tau^p)PH_k$.

At origination, a buyer can finance at most a share ϕ of house value, so the required down payment is

$$(1 - \phi)PH_k. \quad (30)$$

Equivalently, adjusted liquid wealth immediately after purchase must satisfy $\tilde{b} \geq -\phi PH_k$. End-of-period saving for every owner, whether a buyer or an incumbent, faces the same debt floor:

$$b' \geq -\phi PH_k. \quad (31)$$

The down payment governs entry into ownership, while the debt floor governs resources after purchase. The distinction matters because a household can be able to service an owned home and still lack the liquid wealth needed to buy it.

Housing adjustment is costly. A sale of an owned house $h > 0$ incurs the proportional transaction cost ψ^s , so net sale proceeds are

$$\Omega(h) = (1 - \psi^s)Ph. \quad (32)$$

Transaction costs are the computed benchmark's disposal friction.²

3.4 Fertility and Child Aging

Fertility is chosen during the fertile window

$$a \in \{A_f^{\text{start}}, \dots, A_f^{\text{end}}\}.$$

A childless household chooses

$$n' \in \mathcal{N} = \{0, 1, \dots, \bar{n}\}. \quad (33)$$

The option $n' = 0$ means no birth in the current period. A positive choice fixes completed fertility forever and moves its children into the first dependent stage. A household that reaches the end of the fertile window without a positive choice remains childless.

²Capital-gains taxation and stepped-up basis are left for future quantitative work because incorporating them requires tracking each owner's tax basis.

In the computation, $n \in \{0, 1, 2\}$ is a completed-fertility category rather than a literal count of children.³

Let $s = 0$ denote no dependent children and let $s = 1, \dots, S_c$ denote dependent-child stages. After a positive fertility choice, the household enters $s = 1$. Children age stochastically through the stages according to Π^c . While children are at home, $n_s = n$; after maturity, $n_s = 0$. Completed fertility still matters through bequest utility. All children therefore belong to a single cohort and move through the child-age stages together. The model does not yet allow sequential births or children of different ages; adding sequential fertility is the main next extension of this block.

3.5 Household Problem

A household begins the period with liquid assets b , persistent earnings component z , owned housing h , age a , completed fertility n , and child stage s . If the household is childless and in the fertile window, it first observes its fertility taste shocks and chooses family size. Conditional on the current family size, it observes its housing taste shocks and chooses the owned housing position h' . It then chooses consumption c , end-of-period liquid assets b' , and, if renting, current rental services h^R .

The choice h' is made today and becomes the household's owned housing position next period; $h' = 0$ means that the household rents. Rental services h^R , by contrast, are consumed only in the current period and therefore carry no prime.

Before current housing transactions, the bond pays b/q . Liquid wealth after changing housing from h to h' is

$$\tilde{b}(b, h, h') = \frac{b}{q} + \mathbf{1}\{h > 0, h' \neq h\}\Omega(h) - \mathbf{1}\{h' > 0, h' \neq h\}Ph'. \quad (34)$$

When $h' = h$, no house is traded. An owner who changes housing receives the net sale proceeds $\Omega(h)$, while a household choosing a new owned house pays Ph' . The renter and owner problems below are therefore evaluated at $\tilde{b}(b, h, 0)$ and $\tilde{b}(b, h, h')$, respectively.

Let the continuation value after earnings risk and child aging be

$$\bar{V}(b', z, h', a + 1, n, s) = \mathbb{E} [V(b', z', h', a + 1, n, s') \mid z, s, n]. \quad (35)$$

The expectation is over next period's earnings component and child stage. The terminal value is

$$V(b, z, h, J + 1, n, s) = \mathcal{B}(b + Ph, n).$$

Conditional on renting, the branch value is

$$\begin{aligned} V^R(\tilde{b}, z, a, n, s) &= \max_{c, h^R, b'} u(c, h^R, 0; n, s) + \beta \bar{V}(b', z, 0, a + 1, n, s) \\ \text{s.t.} \quad &c + rh^R + b' = \tilde{b} + y(a, z), \\ &0 \leq h^R \leq \bar{h}^R, \quad b' \geq 0. \end{aligned} \quad (36)$$

Conditional on choosing an owned house $h' > 0$ after any adjustment, the branch value is

$$\begin{aligned} V^O(\tilde{b}, z, h', a, n, s) &= \max_{c, b'} u(c, 0, h'; n, s) + \beta \bar{V}(b', z, h', a + 1, n, s) \\ \text{s.t.} \quad &c + (\delta + \tau^p)Ph' + b' = \tilde{b} + y(a, z), \\ &b' \geq -\phi Ph'. \end{aligned} \quad (37)$$

³The category $n = 1$ represents families with one or two children, while $n = 2$ represents families with three or more. The maintained scaling assigns two children to each unit of the index, so the completed-fertility measure matched to the CPS is reported as $2\mathbb{E}[n]$. Preferences, child needs, bequests, and demographic flows are evaluated in index units.

The debt floor is the same for a buyer and an incumbent. Renting offers a continuous choice of space but no borrowing; ownership offers the larger discrete menu and borrowing against the house, at the cost of the initial down payment and future adjustment costs. The housing-position value for a feasible $h' \in \mathcal{H}$ is

$$\tilde{V}^H(h'|b, z, h, a, n, s) = \begin{cases} V^R(\tilde{b}(b, h, 0), z, a, n, s), & h' = 0, \\ V^O(\tilde{b}(b, h, h'), z, h', a, n, s), & h' \in \mathcal{H} \setminus \{0\}. \end{cases} \quad (38)$$

The rent-or-own and owner-size choices have Type-I extreme-value shocks with scale κ_T :

$$\pi^H(h'|b, z, h, a, n, s) = \frac{\exp\{\tilde{V}^H(h'|b, z, h, a, n, s)/\kappa_T\}}{\sum_{\hat{h} \in \mathcal{H}} \exp\{\tilde{V}^H(\hat{h}|b, z, h, a, n, s)/\kappa_T\}}, \quad (39)$$

and the inclusive housing value is

$$V^H(b, z, h, a, n, s) = \kappa_T \log \sum_{\hat{h} \in \mathcal{H}} \exp\{\tilde{V}^H(\hat{h}|b, z, h, a, n, s)/\kappa_T\}. \quad (40)$$

For a childless household in the fertile window, the value of fertility option n' is

$$\tilde{V}^F(n'|b, z, h, a, 0, 0) = V^H(b, z, h, a, n', s(n')), \quad s(n') = \mathbf{1}\{n' > 0\}. \quad (41)$$

With Type-I extreme-value shocks of scale κ_F ,

$$\pi^F(n'|b, z, h, a, 0, 0) = \frac{\exp\{\tilde{V}^F(n'|b, z, h, a, 0, 0)/\kappa_F\}}{\sum_{\hat{n} \in \mathcal{N}} \exp\{\tilde{V}^F(\hat{n}|b, z, h, a, 0, 0)/\kappa_F\}}. \quad (42)$$

The value before the fertility draw is the corresponding inclusive value:

$$V(b, z, h, a, 0, 0) = \kappa_F \log \sum_{\hat{n} \in \mathcal{N}} \exp\{\tilde{V}^F(\hat{n}|b, z, h, a, 0, 0)/\kappa_F\}. \quad (43)$$

Outside the fertile window, or after a positive fertility choice, this stage is degenerate and $V(b, z, h, a, n, s) = V^H(b, z, h, a, n, s)$. The family-size decision therefore incorporates the housing position chosen in the same period. A birth can change current space and tenure immediately, not only the household's continuation value.

3.6 The Entry Problem

At age 18, a potential household chooses whether to live in the modeled housing market or in the outside economy. This choice is made before initial liquid wealth and the persistent earnings component are realized. Conditional on entry, the household draws (b, z) from G_E and begins adult life as a childless renter. Locally born and outside-origin households draw from the same distribution G_E .

The value of entry is therefore

$$W^E(r) = \int V(b, z, 0, 1, 0, 0; r) dG_E(b, z).$$

The outside economy provides value $\bar{W}^E$. With Type-I extreme-value tastes of scale κ_E , the fraction of potential households that enter is

$$q^E(r) = \frac{\exp\{W^E(r)/\kappa_E\}}{\exp\{W^E(r)/\kappa_E\} + \exp\{\bar{W}^E/\kappa_E\}}. \quad (44)$$

Housing costs affect entry through $W^E(r)$: a lower cost of living in the modeled market raises the fraction of potential households that enter.

3.7 Entry Equilibrium and Stationary Scale

Let g denote the beginning-of-period composition of adult households over liquid wealth, the persistent earnings component, inherited owned housing, age, family size, and child age, (b, z, h, a, n, s) , normalized to integrate to one. Let S denote the total adult population.

There are two sources of potential entrants. A fixed mass M comes from the outside economy. In addition, locally born children mature into potential entrants. Let $B_0(g, r)$ denote the flow of mature locally born potential entrants per unit of adult population. The total potential-entrant pool is therefore $M + SB_0(g, r)$.

Let $E_0(g, r)$ denote the inflow of young adults required per unit of adult population to reproduce the stationary composition g —equivalently, the mass of age-one households in g . Stationarity requires the number who enter to equal the number of young adults needed to replace the population:

$$q^E(r) [M + SB_0(g, r)] = SE_0(g, r). \quad (45)$$

The left-hand side is the mass entering from the two potential-entrant pools; the right-hand side is the inflow required to sustain an adult population of size S .

For $M > 0$, a finite positive adult population requires $E_0(g, r) > q^E(r)B_0(g, r)$. Under this condition, rearranging (45) gives

$$S = \frac{q^E(r)M}{E_0(g, r) - q^E(r)B_0(g, r)}. \quad (46)$$

Housing costs affect population scale through the entry probability $q^E(r)$, while fertility affects scale through the future flow $B_0(g, r)$ of locally born potential entrants.

The benchmark normalizes $S^* = 1$.⁴ Given the benchmark composition, I choose $\bar{W}^E$ to deliver the benchmark entry probability $q^{E,*}$ and set

$$M = \frac{E_0^*}{q^{E,*}} - B_0^*.$$

3.8 Stationary Equilibrium

Throughout the equilibrium definition, q , δ , and τ^p are fixed, so $P = r/\rho$.

Definition 1 (Stationary equilibrium). A stationary equilibrium is a rental user cost and asset price (r, P) , value functions and policies, housing and fertility choice probabilities, an entry probability, stationary composition g , adult scale S , and aggregate housing quantities such that:

1. households solve the branch problems (36)–(37), housing choice is given by (39), fertility choice is given by (42), and terminal utility is (26);
2. household choices and saving, together with earnings risk, child and adult aging, death, and an inflow $E_0(g, r) dG_E(b, z)$ of age-one childless renters, define a normalized transition operator $\mathcal{T}(g; r, S)$. Stationarity requires

$$g = \mathcal{T}(g; r, S); \quad (47)$$

3. entry and population scale satisfy (44) and (45), with finite scale given by (46);

⁴Counterfactuals hold $(M, \bar{W}^E, \kappa_E, G_E)$ fixed while entry, population scale, and the housing-market equilibrium adjust.

4. let $\Pi^H(h' \mid b, z, h, a, n, s)$ and $\hat{h}^R(b, z, h, a, n, s)$ denote the housing-position probability and expected rental space conditional on renting, respectively, after integrating over the current family-size choice. Per-unit physical housing demand is

$$\hat{H}^D(g, r) = \int \left[\Pi^H(0 \mid b, z, h, a, n, s) \hat{h}^R(b, z, h, a, n, s) + \sum_{k=1}^K \Pi^H(H_k \mid b, z, h, a, n, s) H_k \right] dg(b, z, h, a, n, s). \quad (48)$$

5. the housing market clears in levels and the user-cost relation holds:

$$S\hat{H}^D(g, r) = H^S(r), \quad r = (q^{-1} - 1 + \delta + \tau^p)P; \quad (49)$$

6. taxes, pensions, and any rebates satisfy the government accounting rule specified for the calibration or counterfactual.

Equilibrium links family formation to housing through two margins. Children change housing demand within a household, and, when locally born children feed the future entrant pool, they also change the scale of the market. Housing supply determines how much of either margin is absorbed by quantities rather than by the user cost.

3.9 Computation

One model period is four years. Period $j = 0, \dots, 16$ begins at age $18 + 4j$; the last interval covers ages 82–85. Retirement begins at 66, and the last period in which a birth can occur begins at 42. Annual parameters are converted before solution:

$$q = (1 + i_{\text{ann}})^{-4}, \quad \beta = \beta_{\text{ann}}^4, \quad \delta = 1 - (1 - \delta_{\text{ann}})^4, \quad \tau^p = 4\tau_{\text{ann}}^p.$$

The property-tax conversion is the undiscounted sum of four annual flow rates. Income and consumption-flow quantities are expressed in four-year units; housing remains in rooms.

The liquid-wealth grid spans $[-12, 30]$, with a dense core on $[-5, 7]$ and a thinner upper tail beyond 15. The five-state Rouwenhorst process has the following grid for the earnings component: $z \in \{0.613, 0.773, 0.974, 1.227, 1.546\}$ and invariant weights $\pi_z = (0.0625, 0.25, 0.375, 0.25, 0.0625)$. One dependent-child stage has an expected duration of 18 years, giving a four-year maturity probability of $2/9$. With deterministic J -period lives, the stationary normalized composition has age-one mass $E_0(g, r) = 1/J$. The balanced pension equals payroll-tax revenue per retiree in the stationary age distribution.

I solve the household problem by backward induction and obtain the stationary distribution by forward iteration. The user cost adjusts until aggregate housing demand equals supply.

4 Calibration

I organize the model parameters into two groups. The first consists of parameters assigned from external evidence, measured directly in the data, or fixed as maintained restrictions. The second consists of parameters estimated internally by matching model moments to their empirical counterparts. I describe the two groups in turn, explain which moments are particularly informative about the internally estimated parameters, and then report the model's fit.

4.1 Parameters Set Outside the Estimation

I first describe the parameters assigned outside the estimation. Table 1 summarizes their values and sources.

Demographics and preferences. A model period is four years. Households enter adult life at age 18, retire at age 66, and live at most through age 85. Survival is certain before retirement and thereafter follows four-year probabilities constructed from the combined male and female survivor counts in the 2023 Social Security period life table. Dependent children remain in the household for an expected 18 years, implying a maturity probability of $2/9$ per period. I set risk aversion to $\sigma = 2$.

Housing finance and transaction costs. Following Greaney et al. (2025), I set the annual real return on liquid wealth to $i_{\text{ann}} = 0.02$, annual housing depreciation to $\delta_{\text{ann}} = 0.011$, the proportional selling cost to $\psi^s = 0.06$, and the labor-income tax rate to $\tau_{\text{pay}} = 0.179$. The return is based on the average ten-year real interest rate, depreciation combines BEA residential-structure depreciation with an adjustment for the land share of house values, and the tax rate comes from OECD data. I set the annual property-tax rate to $\tau_{\text{ann}}^p = 0.01$. Mortgage debt cannot exceed a share $\phi = 0.80$ of the value of the house, so a purchase requires at least 20 percent equity. The benchmark imposes no separate payment-to-income limit.

Income process. I impose a deterministic lifecycle profile of household earnings, e_a , normalized to average one over working ages. Within age, log income follows an annual AR(1) process with persistence $\rho_z = 0.960$, which I sample at the model’s four-year frequency and approximate with a five-state Rouwenhorst chain. This persistence is near the upper end of the PSID-based estimates summarized by Sommer and Sullivan (2018). I set the annual innovation standard deviation to $\sigma_z = 0.0645$, below estimates of gross income risk in that literature. I treat this variance as provisional rather than as an externally validated estimate of income risk.⁵ The quantitative results are conditional on this restriction.

Housing products and supply. Housing is indivisible and available in a limited assortment of types and sizes (Davis and Van Nieuwerburgh, 2015). I represent this lumpiness with five owner-occupied house sizes, $\{H_k\}_{k=1}^5 = \{2, 4, 6, 8, 10\}$ rooms. Renters choose housing continuously up to $\bar{h}^R = 6$ rooms, the 90th percentile of the rental-room distribution in the matched ACS sample. These menus summarize the observed tenure-specific distribution of rooms. I set the housing-supply elasticity to $\eta = 1.75$, following the population-weighted metropolitan estimate in Saiz (2010).

Entry wealth and bequests. Households enter at age 18 with a draw from the PSID net-worth-to-income distribution of childless renters aged 18–24. I impose $\theta_n = 0$, so bequest utility does not vary directly with the number of children. Among PSID households aged 65–75, the median total-wealth-to-income ratio for households with two or more children exceeds that for one-child

⁵With innovation risk in the empirical range, households in the lowest persistent-income states cannot always finance the model’s minimum consumption and housing bundle. This conflict may reflect missing insurance mechanisms—the current model has no means-tested safety net, family transfers, or labor-supply margin—but it may also indicate that the hard minimum requirements implied by the Stone–Geary specification are too restrictive. Future revisions will therefore reconsider both the missing insurance margins and the Stone–Geary preference structure.

households by 0.101. Children affect estate accumulation through their effects on housing needs, tenure, and saving.

Table 1: Parameters set outside the estimation

Parameter	Value	Source or restriction
Model period	4 years	Model timing
Entry, retirement, and maximum ages	18, 66, and 85	Model timing
Post-retirement survival	Age-specific	2023 Social Security period life table
Expected years with dependent children	18 years	Child-maturity restriction
Risk aversion σ	2	
Annual real return i_{ann}	2.0%	Average 10-year real rate, 1962–2024; Greaney et al. (2025)
Annual housing depreciation δ_{ann}	1.1%	BEA structures depreciation, adjusted for the residential land share; Greaney et al. (2025)
Annual property tax τ_{ann}^p	1.0%	Maintained tax rate
Financed share ϕ	0.80	20% down-payment restriction; Davis and Van Nieuwerburgh (2015)
Sale cost ψ^s	6%	Davis and Van Nieuwerburgh (2015) ; Greaney et al. (2025)
Labor-income tax τ_{pay}	17.9%	OECD U.S. average; Greaney et al. (2025)
Income persistence ρ_z	0.960 annually	see text
Income innovation standard deviation σ_z	0.0645 annually	see text
Owner house sizes $\{H_k\}$	$\{2, 4, 6, 8, 10\}$ rooms	Occupied-room distribution in the ACS; see text
Largest rental unit $\bar{h}^R$	6 rooms	Matched ACS rental-size distribution
Housing-supply elasticity η	1.75	Saiz (2010)
Entrant wealth distribution	Wealth-to-income distribution	Childless renters aged 18–24 in the PSID

4.2 Internally Estimated Parameters

I estimate the remaining 14 parameters internally using 15 moments describing completed fertility, childlessness, housing consumption, tenure, wealth accumulation, and aggregate occupied housing. Table 2 reports the estimates. Although the parameters are chosen jointly, I organize the discussion into economic blocks and describe which moments are most informative about each block.

Table 2: Internally estimated parameters

Parameter	Interpretation	Estimate
β_{annual}	Annual discount factor	0.9912
α_c	Consumption weight	0.5912
$\bar{c}_0$	Baseline consumption requirement	1.2596
$\bar{c}_n$	Consumption cost per dependent child	0.3929
$\bar{h}_0$	Baseline housing requirement	0.3915
$\bar{h}_{\text{jump}}$	Housing-need increase with the first child	1.6021
$\bar{h}_n$	Housing requirement per dependent child	0.1644
ψ_{child}	Utility from children	0.1979
κ_F	Dispersion in family-size choices	2.0434
χ_O	Utility premium from owner-occupied housing	1.1133
$\bar{H}$	Housing-supply scale	8.6456
θ_0	Strength of the bequest motive	0.3118
θ_1	Bequest curvature shift	0.3973
κ_T	Dispersion in tenure choices	0.0100

Saving. The annual discount factor β_{annual} governs households' willingness to carry resources across ages. Liquid wealth among young childless renters, the older nonhousing-wealth share, and the late-life estate median are particularly informative about it.

Housing demand, tenure, and supply. I use housing-size differences by family composition to discipline the housing requirements. Mean rooms among childless renters is particularly informative about $\bar{h}_0$, the increase associated with the first child disciplines $\bar{h}_{\text{jump}}$, and the difference between larger and smaller families disciplines $\bar{h}_n$. The owner-renter rooms difference and the share of childless owners in large homes help discipline α_c and χ_O . Ownership rates by age and family status are particularly informative about χ_O and κ_T , while aggregate occupied rooms disciplines the housing-supply scale $\bar{H}$.

Consumption needs and family size. The consumption requirement is $\bar{c}_0 + \bar{c}_n n_s$. Outcomes for childless households, including young liquid wealth, are particularly informative about the baseline requirement $\bar{c}_0$, while fertility and differences in housing and tenure by family composition help discipline the incremental cost $\bar{c}_n$. Completed fertility and childlessness jointly discipline these costs with the direct utility value of children ψ_{child} and the dispersion of family-size choices κ_F . These parameters are estimated jointly because the same family outcomes respond to both the cost and the utility value of children.

Bequests. The bequest parameters θ_0 and θ_1 govern the strength and wealth dependence of the bequest motive. The late-life median estate is particularly informative about θ_0 , while the older nonhousing-wealth share, together with old-age ownership, provides additional information about θ_1 . These moments also respond to saving and tenure choices, so the two bequest parameters are disciplined jointly with those blocks.

4.3 Target Moments and Fit

The target system combines completed fertility and childlessness from the CPS, tenure and housing-size moments by age and family status from the ACS, housing responses to changes in the number of children, and wealth moments from the PSID. Table 3 reports all 15 target moments and their model counterparts.

Table 3: Target moments and model fit

Moment	Data	Model
<i>Family size and family housing</i>		
Completed fertility	1.918	2.034
Childlessness rate	0.188	0.189
Family ownership gap	0.168	0.219
Housing increment with the first child, rooms	0.664	0.681
Rooms gap, 3+ versus 1–2 children, ages 30–55	0.368	0.607
<i>Tenure and housing consumption</i>		
Ownership rate, ages 30–55	0.575	0.658
Renter mean rooms, no children at home, ages 30–55	3.805	3.963
Owner share with rooms ≥ 6 , no children at home, ages 30–55	0.596	0.760
Owner–renter mean rooms gap, no children at home, ages 30–55	2.419	2.323
Ownership rate, ages 25–34	0.341	0.274
Ownership rate, ages 65–75	0.764	0.954
Mean occupied rooms, ages 18–85	5.780	5.673
<i>Wealth and bequests</i>		
Young childless-renter liquid wealth / annual income, ages 25–35	0.179	0.328
Median total estate wealth / annual income, ages 76–84	6.501	6.431
Share with nonhousing wealth $\geq$ annual income, ages 65–75	0.608	0.610

The model closely matches childlessness, the housing increment associated with the first child, renter mean rooms, the owner–renter rooms difference, aggregate occupied rooms, the late-life median estate, and the older nonhousing-wealth share. It produces completed fertility of 2.034, compared with 1.918 in the data. The model overstates the overall ownership rate, the family ownership gap, the share of childless owners in large homes, and the rooms difference between larger and smaller families, while understating ownership at ages 25–34.

The wealth moments reveal a lifecycle tension. The two late-life wealth targets are closely matched, but young childless renters accumulate liquid wealth equal to 0.328 of annual income, compared with 0.179 in the PSID. The estimated annual discount factor of 0.9912 should therefore be read as a joint calibration outcome rather than as a direct estimate of household patience.

The largest miss is old-age ownership: the model produces an ownership rate of 0.954 at ages 65–75, compared with 0.764 in the ACS. The model has no health or long-term-care expense risk and no age-dependent maintenance burden that induces older owners to leave homeownership. Together with transaction costs and a bequest motive that values the house at death, this makes owner housing unusually persistent late in life. The untargeted estate distribution shows a related limitation: the model produces an estate p90–p50 ratio of 1.75, compared with 3.45 in the PSID, and understates nonhousing wealth in the upper part of the late-life distribution. I therefore do not use the current calibration to draw conclusions that depend on the late-life ownership path or portfolio composition.

The calibration remains provisional because the income variance and the late-life exit and portfolio margins require further work. The quantitative exercises that follow focus on family housing needs, tenure, and the down-payment constraint and should be read conditional on these limitations.

5 Lifecycle Housing Patterns

Figure 3 compares the model’s lifecycle housing and child-at-home profiles with the ACS. The model understates homeownership through the mid-thirties and overstates it from the early forties onward. Mean rooms are below the data throughout the lifecycle. The child-at-home profile has the right hump shape but is too delayed and persistent: the model peaks at 0.522 at age 42, whereas the ACS profile peaks at 0.646 at age 41.

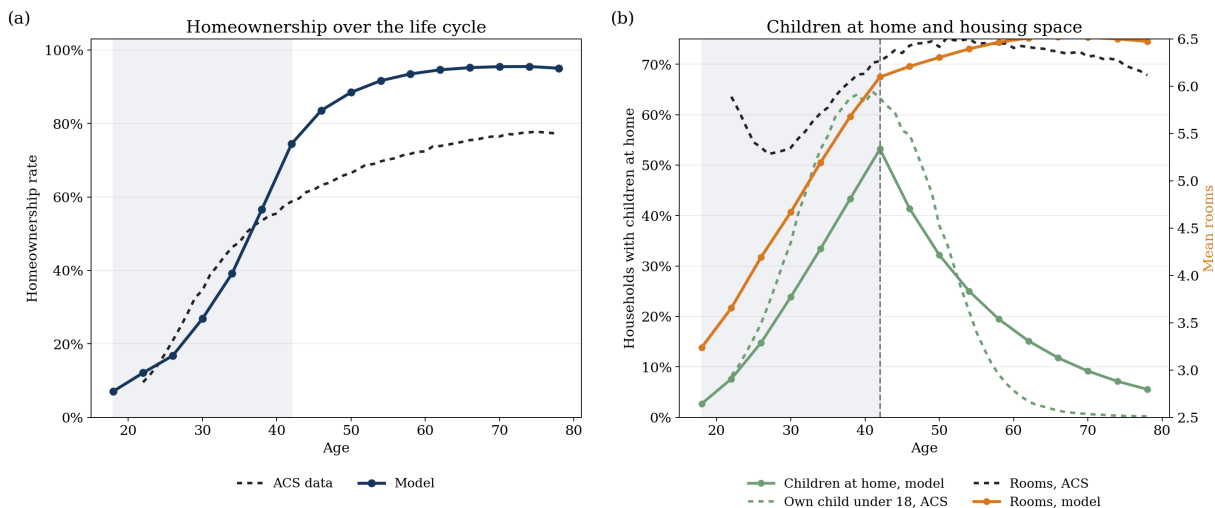


Figure 3: Lifecycle ownership, children at home, and housing consumption. Panel (a) compares model homeownership with the ACS household-head profile. Panel (b) compares the model share of households with dependent children at home with the ACS share of household heads with an own child under age 18, and compares mean rooms in the model and the ACS. The shaded region marks ages 18–42, the family-size-choice window in the model.

At age 38 the model child-at-home share is 18.2 percentage points below the ACS; at age 62 it is 11.7 points above it. Thus the discrepancy is not completed fertility “at” a young age. It is a timing and duration mismatch in the stock of dependent children at home, the object that directly shifts household space requirements in the model.

Figure 4 fixes households at age 42 as childless renters and varies liquid wealth and income. Both choices are strongly nonlinear. Higher-income households select children and larger homes at lower levels of liquid wealth, while the steps in physical rooms reflect movements through the discrete owner ladder.

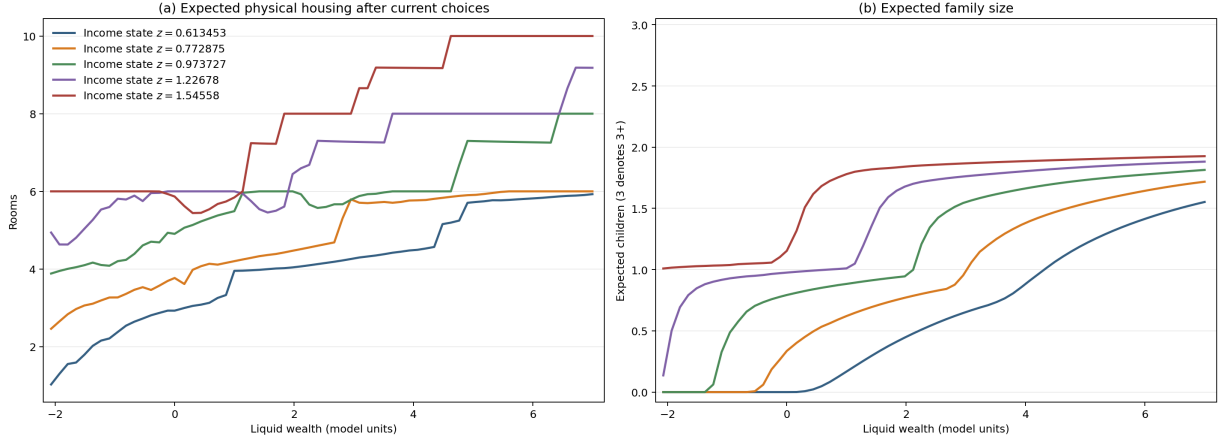


Figure 4: Housing and family-size policies at the end of the fertile window. Panel (a) reports expected physical rooms after integrating over the current family-size choice and the housing choice that follows it. Panel (b) reports expected family size. Each line corresponds to a current income state.

The housing rule is not monotone at every tenure transition: for some states, crossing into ownership reduces physical rooms. This can occur because owner housing is discrete and carries an ownership premium.

6 Quantitative Evidence on the Mechanism

I use the household's model-implied marginal rate of substitution to measure the value of relaxing the rental-size constraint. For a fertile childless renter state i , let (c_i^B, h_i^B) denote consumption and housing on the first positive family-size branch conditional on continuing to rent, and define usable consumption and space by $\tilde{c}_i = c_i^B - \bar{c}(1)$ and $\tilde{h}_i = h_i^B - \bar{h}(1)$. If this branch reaches the rental cap, the excess marginal value of another room is

$$\zeta_i^R = \frac{1 - \alpha_c}{\alpha_c} \frac{\tilde{c}_i}{\tilde{h}_i} - r,$$

where r is the rental user cost per room. Thus $r + \zeta_i^R$ is the household's local marginal willingness to substitute consumption for space. This is a model-implied marginal valuation, not an observed bid or the welfare gain from providing an entire additional room.

Among current fertile renters whose conditional positive-family renter branch reaches the cap, the stationary-mass-weighted mean of ζ_i^R is 0.178 goods units. This is the value above the 0.130 rental user cost. Their total marginal value of another unit of space is 0.307 goods units, or 2.37 times its user cost.

Table 4 compares who owns homes with at least six rooms in the model and in the ACS.

Table 4: Who holds the family-sized owner stock (rooms ≥ 6)

Holder	Data (ACS)	Model
Age 18–29	3.4%	2.4%
Age 30–45	24.8%	16.9%
Age 46–64	43.3%	43.5%
Age 65+	28.5%	37.2%
Held by households 46+	71.8%	80.7%
Held by households with no children at home	54.9%	74.8%

In the ACS, households past the fertile years hold 72 percent of family-sized owner housing, and households with no children at home hold 55 percent. The corresponding model shares are 81 and 75 percent. These quantities describe who holds the stock; whether that allocation is inefficient depends on the holders’ and potential occupants’ marginal values of space.

7 Housing Policy

I use these preliminary policy exercises to ask how relaxing access to family-sized housing affects fertility. The first is a purchase grant for renters with dependent children who buy a home with at least six rooms. The grant covers roughly one-third of the required down payment on an eligible family-sized home. The second exercise combines the same grant with an increase in the annual property-tax rate on owner-occupied housing from 1 to 2 percent. House prices, market entry, and population adjust in both exercises.

The purchase grant raises total births by approximately 2.1 percent. The property-tax-plus-grant package raises total births by approximately 2.7 percent. The corresponding housing-price changes are +0.90 percent and –18.44 percent, respectively. Table 5 summarizes the results.

Table 5: Population-adjusted effects of housing policy

Policy	Fertility effect: Δ total births	Δ housing price
Purchase grant for family-sized homes	+2.1%	+0.90%
Package: higher property tax and purchase grant	+2.7%	–18.44%

The total-birth effects combine changes in family formation among incumbent households with changes in the number of households living in the market. The grant’s approximately 3.4 percent increase in births comes with a 0.66 percent increase in the housing price. The package’s approximately 4.9 percent increase in births comes with an 18.80 percent decline in the housing price.

These exercises are most useful for comparing mechanisms. The model has too little old-age downsizing, omits the capital-gains basis channel, and generates the distorted family-formation timing documented above. Because the model represents one housing market, the total-birth measure cannot distinguish newly created births from births relocated from other markets.

8 Conclusion

Children raise demand for space, but young households face a rental-size limit and, upon buying, a down payment. At the same time, older households hold much of the family-sized owner stock. The analytical model shows how these features raise the effective cost of accommodating children and can misallocate housing across generations. It also shows that capital gains are transition objects: one permanent change can generate appreciation for several successive cohorts, while taxable gains vanish at both positive steady-state endpoints.

The population-adjusted counterfactuals provide the main quantitative result. The purchase grant raises total births by approximately 3.4 percent and the housing price by 0.66 percent. The tax-and-grant package raises total births by approximately 4.9 percent and lowers the housing price by 18.80 percent. The central lesson is that housing policy affects family formation when it changes constrained households' access to family-sized space at the time they make fertility decisions.

A Proofs and computational construction

This appendix derives the conditional household policies, gives sufficient conditions for positive steady states and finite-horizon transition truncations, states conditions under which their limit is an exact transition equilibrium, proves the reallocation and fertility results, and reports a reproducible mixed-tenure transition approximation.

A.1 Old-age problems

An old renter with financial wealth a solves

$$\begin{aligned} V_t^R(a) &= \max_{c^O, h^O, e} \log c^O + \gamma \log h^O + \omega_B \log e \\ \text{s.t.} \quad &c^O + qe + r_t h^O = a + T_t, \quad 0 < h^O \leq \bar{h}^R. \end{aligned} \quad (50)$$

For an old owner, define

$$A = 1 + \gamma + \omega_B, \quad \varrho_t = r_t - \ell_t, \quad X_t = a + (P_t - \ell_t)H + T_t. \quad (51)$$

The owner problem is (5). Both old problems are strictly concave on their feasible sets.

Lemma 1 (Old-age allocations). *Suppose $r_t > 0$, $\varrho_t > 0$, and resources place the household in the log domain. If the renter size limit is slack, then*

$$c^O = \frac{a + T_t}{A}, \quad h^O = \frac{\gamma(a + T_t)}{Ar_t}, \quad e = \frac{\omega_B(a + T_t)}{Aq}. \quad (52)$$

If the renter limit binds, set $Z^R = a + T_t - r_t \bar{h}^R$; then

$$h^O = \bar{h}^R, \quad c^O = \frac{Z^R}{1 + \omega_B}, \quad e = \frac{\omega_B Z^R}{(1 + \omega_B)q}. \quad (53)$$

If both owner constraints are slack, then

$$c^O = \frac{X_t}{A}, \quad h^O = \frac{\gamma X_t}{A\varrho_t}, \quad e = \frac{\omega_B X_t}{Aq}, \quad (54)$$

and the two strict slackness conditions are

$$\frac{\gamma X_t}{A\varrho_t} < H, \quad \omega_B \varrho_t > \gamma q P_{t+1}. \quad (55)$$

Proof. For a generic interior budget $c^O + qe + p_h h^O = X$, the three first-order conditions imply $p_h h^O = \gamma c^O$ and $qe = \omega_B c^O$. The budget gives $A c^O = X$. Substituting $p_h = r_t$ for renters and $p_h = \varrho_t$ for owners gives (52) and (54). If renter housing is fixed at $\bar{h}^R$, strict concavity allocates the remaining resources between consumption and the estate in shares $1 : \omega_B$, which gives (53). Substitution of the interior owner policy into $h^O < H$ and $e > P_{t+1} h^O$ yields (55). $\square$

The remaining owner active sets are also explicit. If only estate composition binds, let $d_t = \varrho_t + q P_{t+1}$; then

$$c^O = \frac{X_t}{A}, \quad h^O = \frac{(\gamma + \omega_B) X_t}{A d_t}, \quad e = P_{t+1} h^O. \quad (56)$$

If only retention binds, set $Z_t = X_t - \varrho_t H$; then

$$h^O = H, \quad c^O = \frac{Z_t}{1 + \omega_B}, \quad e = \frac{\omega_B Z_t}{(1 + \omega_B)q}. \quad (57)$$

If both constraints bind,

$$h^O = H, \quad e = P_{t+1}H, \quad c^O = X_t - (\varrho_t + qP_{t+1})H. \quad (58)$$

The estate-only row requires $\gamma q P_{t+1} \geq \omega_B \varrho_t$ and $h^O < H$. The retention-only row requires

$$\omega_B Z_t > (1 + \omega_B)qP_{t+1}H, \quad \gamma Z_t \geq (1 + \omega_B)\varrho_t H. \quad (59)$$

For the both-binding row, the multipliers are nonnegative exactly when

$$\frac{q}{c^O} \geq \frac{\omega_B}{P_{t+1}H}, \quad \frac{\gamma + \omega_B}{H} \geq \frac{\varrho_t + qP_{t+1}}{c^O}. \quad (60)$$

These rows and complementary slackness exhaust the strictly concave owner problem.

In the interior owner region, the indirect value and relevant envelopes are

$$V_t^O(a, H) = A \log X_t - A \log A + \gamma \log \gamma - \gamma \log \varrho_t + \omega_B \log \omega_B - \omega_B \log q, \quad (61)$$

$$V_{a,t}^O = \frac{A}{X_t} = \frac{1}{c_t^O}, \quad V_{H,t}^O = (P_t - \ell_t)V_{a,t}^O. \quad (62)$$

A.2 Young household solutions and tenure

The renter and owner problems are those in the main text. Let λ_t^m be the current-budget multiplier, η_t^m the multiplier on the tenure-specific housing limit, v_t^m the multiplier on $a' \geq 0$, and μ_t the owner down-payment multiplier. The common consumption and fertility conditions are

$$\lambda_t^m = \frac{1}{c_t^m}, \quad (63)$$

$$\frac{\vartheta_t}{n_t^m} = \lambda_t^m \chi + \frac{\alpha \kappa}{h_t^m - \kappa n_t^m}. \quad (64)$$

The saving conditions are

$$\lambda_t^R = \beta R V_{a,t+1}^R + v_t^R, \quad (65)$$

$$\lambda_t^O = \beta R V_{a,t+1}^O + v_t^O, \quad v_t^m a_t^m = 0. \quad (66)$$

When saving and next-period old choices are interior, the housing conditions reduce to

$$\frac{\alpha c_t^m}{h_t^m - \kappa n_t^m} = r_t^m + \zeta_t^m, \quad \zeta_t^R = \frac{\eta_t^R}{\lambda_t^R}, \quad \zeta_t^O = \frac{\mu_t}{\lambda_t^O} (1 - \phi) P_t + \frac{\eta_t^O}{\lambda_t^O}. \quad (67)$$

This proves (12). If the owner-size limit is slack, the second term in ζ_t^O is zero and the access wedge is purely financial.

Proposition 3 (Lifetime reduction and conditional policies). *Suppose young saving and next-period old constraints are interior. Then the conditional lifetime budget is (8). If the housing limit is also slack, the policies are (9) and saving is*

$$a_t'^R = kc_t^R - qT_{t+1}, \quad (68)$$

$$a_t'^O = kc_t^O - qT_{t+1} - [q(P_{t+1} - \ell_{t+1}) - \phi P_t]h_t^O. \quad (69)$$

If the effective housing limit binds, $h_t^m = \hat{h}_{it}^m$ and fertility is the unique root of (11) in

$$0 < n < \min \left\{ \frac{\hat{h}_{it}^m}{\kappa}, \frac{\mathcal{W}_{it} - r_t^m \hat{h}_{it}^m}{\chi} \right\}. \quad (70)$$

Proof. The saving Euler equation and Lemma 1 imply $c_{t+1}^O = \beta R c_t$ and therefore $qX_{t+1} = kc_t$. For an owner,

$$X_{t+1} = Ra_t' - R\phi P_t h_t + (P_{t+1} - \ell_{t+1})h_t + T_{t+1}. \quad (71)$$

Substitute this expression and the current owner budget into $qX_{t+1} = kc_t$. The mortgage inflow and repayment cancel, leaving $(1+k)c_t + \chi n_t + [r_t + q\ell_{t+1}]h_t = \mathcal{W}_{it}$. The renter derivation gives the same identity with $r_t^R = r_t$.

Let $s = h - \kappa n$. When housing is unconstrained, (67) gives $s = \alpha c / r_t^m$ and (64) gives $n = \vartheta_t c / (\chi + \kappa r_t^m)$. Substitution into the lifetime budget gives $\mathcal{W}_{it} = (1+k+\alpha+\vartheta_t)c$, proving (9). The identity $qX_{t+1} = kc_t$ gives (68)–(69).

When the limit binds, substitute $h = \hat{h}$ and $c = (\mathcal{W} - r_t^m \hat{h} - \chi n) / (1+k)$ into the fertility condition. This gives (11). Its left side tends to $+\infty$ at the lower boundary of (70), tends to $-\infty$ at the upper boundary, and has derivative

$$-\frac{\vartheta_t}{n^2} - \frac{\alpha \kappa^2}{(\hat{h} - \kappa n)^2} - \frac{(1+k)\chi^2}{(\mathcal{W} - r_t^m \hat{h} - \chi n)^2} < 0. \quad (72)$$

The root therefore exists and is unique. $\square$

If saving is zero or a different old-age active set applies, the lifetime reduction is replaced by the KKT system, but strict concavity still makes each conditional tenure policy unique. The paper results state their active-set conditions explicitly rather than extending closed forms through those corners.

With the type-I extreme-value shocks in (6), the conditional probability of ownership is strictly between zero and one whenever the two deterministic values are finite. Because the logit formula and the conditional policies are continuous, aggregate choices are continuous across the deterministic rent–own indifference locus. An atomless continuous tenure taste distribution gives the same regularity without the logit specialization.

A.3 Aggregation and positive steady states

Let F denote the distribution of observed young types $i = (y_i, b_i)$. The old-state distribution is over financial wealth a , inherited owner housing H , and tenure. For any bounded measurable test function ψ , its law of motion is

$$\int \psi dG_{t+1} = \int \left[(1 - \pi_t^O(i)) \psi(Ra_t'^R(i), 0, R) \right]$$

$$+ \pi_t^O(i) \psi(Ra_t'^O(i) - R\phi P_t h_t^O(i), h_t^O(i), O) \Big] dF(i). \quad (73)$$

Average fertility and young housing integrate the corresponding conditional policies with the same weights. Average old housing integrates the old policies over G_t . Average taxable sales are

$$\bar{s}_t^O = \int \mathbf{1}\{m = O\} [H - h_t^O(a, H, O)] dG_t. \quad (74)$$

Equations (13)–(14), the cohort laws, and (73) close the economy.

Proof of Proposition 1. In a steady state, $O_{t+1} = Y_t$ implies $O^* = Y^*$. Stationarity of the young cohort then implies $1 = \nu \mathcal{N}(P^*, T^*; \vartheta)$. A constant real price makes $\ell^* = 0$ directly from (3). Housing clearing gives $Y^* = \bar{H}/\mathcal{H}$. The government budget becomes $2Y^*T^* = \tau^p P^* \bar{H}$. Eliminating Y^* gives the second equation in (15). Reversing these steps establishes sufficiency once the household policies and stationary distribution are constructed. $\square$

The following result gives checkable sufficient conditions for existence. It also makes clear that a positive steady state is not guaranteed for arbitrary primitives.

Proposition 4 (Steady-state existence and local uniqueness). *Let $\mathcal{D} = [\underline{P}, \bar{P}] \times [\underline{T}, \bar{T}] \subset \mathbb{R}_{++}^2$ be a rectangle on which conditional policies are feasible, relevant user costs are positive, and $\mathcal{N}$ and $\mathcal{H}$ are continuous. Define*

$$\Psi(P, T) = \frac{\tau^p P}{2} \mathcal{H}(P, T; \vartheta). \quad (75)$$

Suppose:

1. $\Psi(P, \cdot)$ maps $[\underline{T}, \bar{T}]$ into itself for every P , and $\sup_{\mathcal{D}} |\Psi_T| < 1$;
2. writing $T = \mathcal{T}(P)$ for its unique fixed point,

$$[\nu \mathcal{N}(\underline{P}, \mathcal{T}(\underline{P}); \vartheta) - 1][\nu \mathcal{N}(\bar{P}, \mathcal{T}(\bar{P}); \vartheta) - 1] < 0. \quad (76)$$

Then a positive steady state exists in $\mathcal{D}$. It is locally unique if

$$J^* = \begin{bmatrix} \nu \mathcal{N}_P & \nu \mathcal{N}_T \\ -\frac{\tau^p}{2}(\mathcal{H} + P\mathcal{H}_P) & 1 - \frac{\tau^p P}{2} \mathcal{H}_T \end{bmatrix}_{(P^*, T^*, \vartheta)} \quad (77)$$

is nonsingular.

Proof. The contraction theorem gives a unique fiscal fixed point $\mathcal{T}(P)$ for each P . Uniform contraction and continuity of Ψ imply continuity of $\mathcal{T}$. The intermediate value theorem applied to the replacement residual along this locus gives a price P^* satisfying the first steady-state equation. Its fiscal fixed point $T^* = \mathcal{T}(P^*)$ satisfies the second. Local uniqueness follows from the implicit-function theorem when J^* is nonsingular. $\square$

The first condition can be checked from bounds on primitives and policies. For example, if $0 < \underline{\mathcal{H}} \leq \mathcal{H} \leq \bar{\mathcal{H}}$ on $\mathcal{D}$, it is enough to choose transfer bounds such that

$$\underline{T} \leq \frac{\tau^p \underline{P}}{2} \underline{\mathcal{H}}, \quad \bar{T} \geq \frac{\tau^p \bar{P}}{2} \bar{\mathcal{H}}, \quad \frac{\tau^p \bar{P}}{2} \sup_{\mathcal{D}} |\mathcal{H}_T| < 1. \quad (78)$$

The replacement boundary condition says that the low- and high-price faces place average fertility on opposite sides of $1/\nu$ after closing the fiscal account. It is an inequality in the unique household policies, and hence is directly verifiable from primitives rather than an assumed equilibrium.

If J^* is nonsingular, the response to a permanent change in ϑ is

$$\begin{bmatrix} P_\vartheta^* \\ T_\vartheta^* \end{bmatrix} = -(J^*)^{-1} \begin{bmatrix} \nu \mathcal{N}_\vartheta \\ -\frac{\tau^p P^*}{2} \mathcal{H}_\vartheta \end{bmatrix}. \quad (79)$$

The associated population-scale response is

$$\frac{d \log Y^*}{d \vartheta} = -\frac{\mathcal{H}_P P_\vartheta^* + \mathcal{H}_T T_\vartheta^* + \mathcal{H}_\vartheta}{\mathcal{H}}. \quad (80)$$

There is no general sign without restrictions on the demand and tenure elasticities.

A.4 Transition equilibrium

Suppose $\vartheta_t = \vartheta^0$ for $t < 0$ and $\vartheta_t = \vartheta^1$ for $t \geq 0$, and both values support regular positive steady states $\mathcal{E}^0$ and $\mathcal{E}^1$. The date-zero state is the old steady state: $S_0 = (Y^0, Y^0, G^0, P^0)$. In particular, $P_{-1} = P^0$ is the basis of date-zero old owners. No additional initialization shock is present.

The gains rule is equivalently the complementarity system

$$\ell_t \geq 0, \quad \ell_t \geq \tau^g(P_t - P_{t-1}), \quad \ell_t[\ell_t - \tau^g(P_t - P_{t-1})] = 0. \quad (81)$$

It is continuous but kinked. Within an appreciating regime its price derivative is τ^g ; within a non-appreciating regime it is zero.

Fix a terminal date J and use the new steady-state prices, transfers, and continuation values from $J+1$ onward. The state generated at $J+1$ is not separately forced to equal the new steady-state state. The resulting object is therefore a terminally closed truncation, not necessarily an exact finite prefix of an infinite-horizon transition. Let z_J stack $(\log P_0, \log T_0, \dots, \log P_J, \log T_J)$, with residuals interleaved in the same order. Recursively applying household policies, the distribution law, and cohort laws turns (17) into a finite system

$$\mathbf{R}_J(z_J; \varepsilon) = 0, \quad (82)$$

where ε scales the permanent change from ϑ^0 to ϑ^1 and the terminal steady state also varies with ε .

Because the constant path lies at the gains kink, local analysis is directional. Choose a candidate appreciation pattern $d = (d_0, \dots, d_{J+1}) \in \{0, 1\}^{J+2}$ and define the smooth branch

$$\ell_t^d = \tau^g d_t (P_t - P_{t-1}). \quad (83)$$

It is a branch of the statutory rule when $P_t - P_{t-1} > 0$ for $d_t = 1$ and $P_t - P_{t-1} < 0$ for $d_t = 0$. Let $\mathbf{R}_J^d$ denote the residual system evaluated on this branch.

Proposition 5 (Directional local existence of a truncation). *Suppose the old steady state is regular, the household active sets are strict, and the branch Jacobian $\partial \mathbf{R}_J^d / \partial z_J$ is nonsingular at $\varepsilon = 0$. Suppose also that the directional solution has*

$$\left. \frac{d(P_t - P_{t-1})}{d\varepsilon} \right|_0 > 0 \text{ if } d_t = 1, \quad \left. \frac{d(P_t - P_{t-1})}{d\varepsilon} \right|_0 < 0 \text{ if } d_t = 0. \quad (84)$$

Then, for each fixed J , a unique branch-consistent terminally closed truncation exists for all sufficiently small positive shocks, varies continuously differentiably with the shock, and satisfies the statutory gains rule on dates $0, \dots, J$.

Proof. Steady-state regularity and Proposition 4 give a continuously differentiable terminal steady state for small ε . Strict active sets, positive log domains, the logit tenure share, and (83) make $\mathbf{R}_J^d$ continuously differentiable in (z_J, ε) . At $\varepsilon = 0$, the constant old-steady-state path is a root on every branch. Nonsingularity and the implicit-function theorem give a unique differentiable branch solution. The strict directional signs in (84) persist for sufficiently small positive ε and validate every selected branch. Therefore $\ell_t^d = \tau^g[P_t - P_{t-1}]_+$ and the branch solution solves the truncated statutory system. $\square$

If a directional price difference is zero, ordinary branch analysis is inconclusive at that date; one must use higher-order terms, a generalized Jacobian, or the continuous global existence result below. Thus the local proposition does not hide the steady-state kink inside a differentiability assumption.

A sufficient derivative condition for nonsingularity is strict row diagonal dominance,

$$\left| \frac{\partial R_{J,k}}{\partial z_{J,k}} \right| > \sum_{\ell \neq k} \left| \frac{\partial R_{J,k}}{\partial z_{J,\ell}} \right| \quad \text{for every } k. \quad (85)$$

All terms are derivatives of the explicit household policies and state laws. Diagonal dominance is sufficient, not necessary; the numerical construction below checks the singular values of the full stacked Jacobian directly.

The next result provides a global existence criterion for the truncated system that does not presuppose a clearing self-map.

Proposition 6 (Boundary-sign existence of a truncation). *Let $\mathcal{B}_J = \prod_{k=1}^{2(J+1)} [\underline{z}_k, \bar{z}_k]$ be a box on which resources and user costs are positive, household policies are well-defined, and the induced states remain in a compact set. Suppose the stacked residual is continuous on $\mathcal{B}_J$ and, for every coordinate k , either*

$$R_{J,k}(\underline{z}_k, z_{-k}) \leq 0 \leq R_{J,k}(\bar{z}_k, z_{-k}) \quad \text{for every } z_{-k}, \quad (86)$$

or the two inequalities are reversed. Then a terminally closed truncation exists in $\mathcal{B}_J$.

Proof. The Poincaré–Miranda theorem applied to the continuous vector $\mathbf{R}_J$ gives a zero in the box. The max operator in (81) preserves continuity, so the argument allows appreciation kinks even though ordinary derivatives fail there. $\square$

The boundary signs have an economic interpretation. The price faces require housing excess demand and excess supply at the respective bounds, after all forward-looking choices and inherited states are evaluated. The transfer faces require revenues to exceed rebates at the lower bound and rebates to exceed revenues at the upper bound. These inequalities are sufficient and must be verified for the parameterization; the model does not guarantee them for arbitrary primitives.

An infinite-horizon result requires an additional limiting argument.

Proposition 7 (Limit of terminally closed truncations). *Let $x_t^J = (P_t^J, T_t^J, S_t^J)$ denote a root of the horizon- J truncated system. Suppose these roots lie in a common compact set, their induced state distributions are tight, and there is a sequence $J_k \rightarrow \infty$ along which $x_t^{J_k}$ converges for every fixed*

date t (with weak convergence for the distribution component). Suppose also that the roots satisfy the uniform-tail condition

$$\lim_{L \rightarrow \infty} \sup_{J \geq L} \sup_{L \leq t \leq J} d(x_t^J, \mathcal{E}^1) = 0. \quad (87)$$

If household policies and state laws are continuous in these topologies, the pointwise limit is an infinite-horizon transition equilibrium converging to $\mathcal{E}^1$.

Proof. Fix any date t . For all sufficiently large k , the horizon- J_k root satisfies the date- t household, market-clearing, government, cohort, and distribution equations. Passing to the limit preserves these equations by continuity. Because t was arbitrary, the pointwise limit satisfies every finite-date equilibrium condition. Equation (87) implies convergence of that limit to $\mathcal{E}^1$. $\square$

The horizon and terminal checks in the numerical construction are evidence compatible with this convergence argument, but two horizons cannot verify the uniform-tail condition and hence do not prove an infinite-horizon equilibrium.

Corollary 1 (One permanent shock and multiple taxable gains). *If the equilibrium following the single permanent shock has $P_s > P_{s-1}$ for $s = t, \dots, t + L$, then $\ell_s > 0$ for all $L + 1$ dates.*

Proof. Equation (3) gives $\ell_s = \tau^g(P_s - P_{s-1}) > 0$ on each appreciating date. The two-period life determines the one-period basis P_{s-1} of cohort s ; it places no restriction on the length of the aggregate price transition. $\square$

A.5 Conditional reallocation

For a constrained young owner, (67) gives the current-goods marginal value

$$MV_{it}^Y = \frac{\alpha c_{it}^O}{h_{it}^O - \kappa n_{it}^O} = r_t + q\ell_{t+1} + \zeta_{it}^{O,F}. \quad (88)$$

For an old incumbent at an interior retention choice, the housing first-order condition from (5) gives

$$MV_{jt}^O = \frac{\gamma c_{jt}^O}{h_{jt}^O} = r_t - \ell_t. \quad (89)$$

Their difference is the left side of (19) before transaction costs.

Assumption 1 (Two-date transfer ledger). For an infinitesimal transfer of housing from incumbent j to buyer i , the allocation authority can: (i) transfer both title and occupancy at date t ; (ii) provide pledgeable bridge finance without relaxing other households' constraints; (iii) make household-specific lump-sum transfers at t and $t + 1$; (iv) reclaim the incremental equal gains-tax rebates on the transferred unit at both dates; and (v) ensure the buyer's next-period retention constraint is slack, so the marginal unit is resold at $t + 1$. Fertility, tenure, cohort masses, and all unaffected allocations are held fixed.

Proof of Proposition 2. Transfer $\epsilon > 0$ units from the old incumbent to the young buyer. At the baseline allocation, the young buyer is willing to surrender $MV_{it}^Y \epsilon + o(\epsilon)$ current goods without losing utility, while the old incumbent requires $MV_{jt}^O \epsilon + o(\epsilon)$ current goods to be compensated. The bridge makes the transfer feasible despite the down-payment constraint. The old sale realizes

$\ell_t \epsilon$ of date- t revenue; the buyer's marginal resale realizes $\ell_{t+1} \epsilon$ at $t + 1$. Under Assumption 1, the authority reclaims the associated rebates, whose date- t value is $\ell_t \epsilon + q \ell_{t+1} \epsilon$.

After paying the real resource cost $K_{ijt} \epsilon$, the first-order amount available for compensation is

$$[MV_{it}^Y - MV_{jt}^O - K_{ijt}] \epsilon = [\zeta_{it}^{O,F} + \ell_t + q \ell_{t+1} - K_{ijt}] \epsilon. \quad (90)$$

If the bracket is strictly positive, transfers can leave both affected households weakly above baseline and one strictly above it. Smoothness and the strict first-order inequality imply the same result for a sufficiently small finite ϵ . At a steady state, $\ell_t = \ell_{t+1} = 0$, giving the stated condition $\zeta_i^{O,F} > K_{ij}$. $\square$

The transfer ledger is part of the proposition, not an institutional detail. Without pledgeable bridge finance or two-date transfers, the marginal-value gap remains a diagnostic of misallocation but need not be implementable as a Pareto improvement. The proposition also fixes population. Policies that change fertility compare economies with different sets of people, for which a social criterion must be stated explicitly; see Golosov et al. (2007) and Pérez-Nievas et al. (2019). The paper therefore reports the conditional allocation result rather than a global welfare theorem.

A.6 Housing access and fertility

Suppose $h = \hat{h}$ binds and fertility remains interior. Define

$$D = \mathcal{W}_{it} - r_t^m \hat{h} - \chi n, \quad S = \hat{h} - \kappa n, \quad c = \frac{D}{1+k}. \quad (91)$$

Derivation of (20) and (21). Let the left side of (11) be $F(n, \hat{h})$. Direct differentiation gives

$$F_n = -\frac{\vartheta_t}{n^2} - \frac{\alpha \kappa^2}{S^2} - \frac{(1+k)\chi^2}{D^2} < 0, \quad (92)$$

$$F_{\hat{h}} = \frac{\alpha \kappa}{S^2} - \frac{(1+k)\chi r_t^m}{D^2}. \quad (93)$$

The implicit-function theorem gives $dn/d\hat{h} = -F_{\hat{h}}/F_n$, which is (20). At a constrained allocation, (67) and $c = D/(1+k)$ imply

$$\frac{D}{S} = \frac{(1+k)(r_t^m + \zeta_{it}^m)}{\alpha}. \quad (94)$$

Substitute this ratio into $F_{\hat{h}} \geq 0$ and rearrange to obtain (21). $\square$

A primitive sufficient condition, stronger than the exact condition, is

$$\chi \leq \frac{(1+k)\kappa r_t^m}{\alpha}, \quad (95)$$

because $\zeta_{it}^m \geq 0$. When the down-payment constraint alone binds,

$$\hat{h}_{it}^O = \frac{b_i}{(1-\phi)P_t}, \quad \frac{\partial \hat{h}^O}{\partial b_i} = \frac{1}{(1-\phi)P_t}, \quad \frac{\partial \hat{h}^O}{\partial \phi} = \frac{b_i}{(1-\phi)^2 P_t}. \quad (96)$$

These expressions give the direct fertility response to a local relaxation. A funded policy also moves prices, transfers, lifetime wealth, and tenure, so its aggregate response requires the equilibrium system.

A.7 Solved mixed-tenure transition approximation

The numerical construction specializes observed types to common income and liquid wealth while retaining the iid tenure tastes in (6). It gives exact steady-state equilibria and a terminally closed transition approximation; it is neither a calibration nor a numerical proof of the infinite-horizon limiting conditions. Table 6 lists the parameters.

Table 6: Parameters of the mixed-tenure construction

Object	Parameter	Value	Object	Parameter	Value
Bond price	q	0.50	Mortgage share	ϕ	0.80
Property tax	τ^P	0.05	Gains-tax rate	τ^g	0.20
Space weight	α	0.40	Space per child	κ	0.50
Goods per child	χ	0.15	Young discount	β	0.40
Old housing weight	γ	0.30	Estate weight	ω_B	0.40
Income	y	1.00	Liquid wealth	b	0.06
Household formation	ν	2.00	Housing stock	$\bar{H}$	2.00
Rental limit	$\bar{h}^R$	0.45	Owner limit	$\bar{h}^O$	2.00
Owner intercept	$\bar{\xi}$	-0.15	Taste scale	κ_T	0.12

At date zero, ϑ rises permanently from 0.35 to 0.50. Solving (15) gives Table 7.

Table 7: Positive steady-state endpoints

Object	Initial steady state	Final steady state
Fertility preference ϑ	0.3500	0.5000
Average fertility $\bar{n}$	0.5000	0.5000
House price P	0.3405	0.4829
Per-household transfer T	0.01154	0.01189
Young owner share	0.7419	0.5096
Renter fertility	0.3511	0.4366
Owner fertility	0.5518	0.5610
Young renter housing	0.4500	0.4500
Young owner housing	0.8810	0.6212
Old renter housing	0.4500	0.4500
Old owner housing	0.6327	0.4457
Young and old cohort mass	1.4757	2.0304

Average fertility is $1/\nu = 0.5$ at both endpoints, but conditional fertility, tenure, housing, prices, and population scale differ. The determinants of the two steady-state Jacobians, computed in $(\log P, \log T)$, are -0.00913 and -0.00838 , so both roots are locally regular.

The horizon-28 truncation uses the final steady-state continuation values, prices, and transfers from date 29 onward and solves all date-by-date housing and government equations through date 28 jointly. The state generated at date 29 is recorded rather than imposed. On impact, average fertility rises to 0.6248; the young cohort then expands with the OLG lag. The price moves from 0.3405 before the shock to 0.3772 on impact, continues to 0.4994 by date 3, and then approaches 0.4829 with damped oscillations. Realized gains taxes are materially positive on each of dates 0 through 3, even though the economy receives only the date-zero permanent preference shock.

The computed truncation has both tenures on every displayed date: the owner share lies between

0.4704 and 0.7877. Minimum renter and owner saving are 0.3530 and 0.4039. The unconstrained young renter and owner housing policies exceed their respective limits by at least 0.5081 and 0.3574, so both young limits bind strictly. The owner-size limit has slack of at least 1.2046, so the owner wedge is purely financial. The old renter cap has a minimum shadow wedge of 0.00676. Old owners sell at least 0.1663 units, their incumbent user cost is at least 0.1770, and their estate-composition slack is at least 0.0228; hence the old-owner interior formulas apply strictly throughout.

The maximum absolute scaled housing or government residual is 2.07×10^{-10} . The maximum terminal-state discrepancy is 1.10×10^{-8} . Replacing horizon 28 by horizon 24 changes the first nine dates in prices, transfers, masses, tenure, fertility, and housing by at most 1.84×10^{-15} . At the computed horizon-28 root, freezing the realized appreciation pattern gives a smallest stacked-Jacobian singular value of 0.401 and a condition number of 48.6. These are conditioning diagnostics for that finite root, not a verification of the directional derivative assumptions at a zero-size shock. Finally, the analytical fertility derivative is 0.6490218128; a centered finite difference gives 0.6490218128, with absolute error 8.0×10^{-12} .

The construction is generated by `code/model/tools/build_simplified_olg_mixed_tenure_theory.py`. It writes the steady states, full truncated path, verification ledger, and the two figures used in the paper. These files make the finite-root and active-set claims reproducible; they do not replace the limiting conditions in Proposition 7.

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